

Alberta AI Data Centres: Environmental and Regulatory Concerns

Category: Environmental, Water Table, Heat & Wildlife Impacts | Evidence ID: #186

Archived: 2026-10-04 20:19

- **Original URL:** <https://elc.ab.ca/post-library/are-we-being-smart-about-ai-data-centres/>
 - **Research Category:** Environmental, Water Table, Heat & Wildlife Impacts
 - **Archival File:** 186_elc_ab_ca_post-library_are-we-being-smart-about-ai-data.pdf
 - **Extraction Status:** Full Web Transcript PDF
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Source Content

Early indications are that environmental assessments are not being required, limiting our data on data centres. Both federally and provincially, the government has been supportive of Artificial Intelligence (AI) data centre development. Recent legislative and policy changes have been made to facilitate the establishment of AI data centres in Alberta. However, data centres have repeatedly been met with local opposition, see for example: - the proposed Wonder Valley project (Hannah Kavanagh, “First Nation calls for federal review of Wonder Valley AI project” (June 4, 2026); Jody MacPherson, “O’Leary’s Data Centre Will Be One of Earth’s Largest Heat Sources, Physicist Warns” (June 5, 2026); Eagle Andersen, “Northern Albertans raise concerns about Wonder Valley AI data centre at packed community hall” (June 5, 2026); Emma Zhao, “First Nation in court to challenge proposed Wonder Valley AI data centre project in northern Alberta” (June 10, 2026) - the proposed Synapse project in Olds (Jody MacPherson, “Hundreds of Rural Albertans Line Up to Battle Data Centre Goliath” (May 29, 2026); Rukhsar Ali, “Olds power plant, data centre application paused as Alberta regulator seeks answers (June 17, 2026); Jody MacPherson, “Application Delayed as 680 Local Residents Gain Standing on Olds, Alberta Data Centre Proposal” (June 19, 2026)); and - the Kinetikor proposal in Rocky View County (Luc Rinaldi, “The Farmers Who Fought a Data Centre – and Won” (May 25, 2026)). The opposition to data centres is not unique to Alberta, see for example the proposed O’Leary AI data centre in Utah (Jody MacPherson, “Opposition Mounts as O’Leary’s Data Centre Approved in Drought-Stricken Utah” (May 14, 2026); Sarah Petz, “Kevin O’Leary wants to build a massive AI data centre in Utah. Some residents aren’t happy” (May 23, 2026); and Jody MacPherson, “O’Leary’s AI Data Centre Projects Face Backlash on Both Sides of The Boarder” (June 3, 2026)). So what exactly are data centres? How are they regulated? And what changes are needed to move forward with data centres without creating environmental damage? Processes and safeguards are needed to evaluate and regulate the tremendous use of power, water and space by data centres. This includes environmental assessment, and issuance of approvals and licenses specific to data centres. What are data centres? The Data Centre Regulation defines data centres as (section 1(1)): a facility or facilities, or a proposed facility or facilities, the purpose of which is to house and operate computing equipment for applications including data hosting and cloud computing, digital asset mining, artificial intelligence and machine learning and digital services and content delivery. [1] Data centres – especially those dedicated to AI and cryptocurrency – require tremendous amounts of power, water and space for their operations. Government support for data centres In 2024, the Government of Alberta (GOA) issued its AI Data Centre Strategy. This strategy centres on three pillars – power capacity,

sustainable cooling and economic growth – with the goal of making Alberta the North American leader in AI driven data centres. In support of this goal, the GOA has created an online portal and a concierge program to facilitate data centre projects. Alberta’s goal to become an AI data centre leader is reflected in the November 2025 Memorandum of Understanding between Canada and Alberta (MOU).[2] One of the stated objectives of the MOU is: Increasing electrical generation for consumer and industrial use on Alberta’s electricity grid, including meeting the needs of AI data centres, while simultaneously reaching net-zero greenhouse gas emissions for the electricity sector by 2050. The MOU goes on to highlight intended projects including “construction of thousands of megawatts of AI computing power” and “construction of large transmission interties with British Columbia and Saskatchewan to strengthen the ability of the western power markets to supply low carbon power to ... data centres” among other industries. Under the MOU, it was agreed that the federal government would provide feedback on Alberta’s policy framework for AI data centres which is to be finalized by Alberta on or before July 1, 2026. Although there have been recent legislative changes, as discussed below, it is not clear that Alberta’s policy framework has been completed. Despite this, data centre applications in the province are forging ahead. How are data centres regulated? The regulatory and policy framework for data centres is rapidly developing. Currently, any data centre-specific legislation and policy is focused on ensuring an adequate supply of electricity to meet data centre requirements without overwhelming the province’s electrical grid. Other aspects of data centre projects are dealt with via existing municipal permitting processes under the Municipal Government Act, approvals and licensing under the Water Act, and approvals under the Environmental Protection and Enhancement Act. Greenhouse gas emissions from co-located power plants may be subject to requirements under the Emissions Management and Climate Resilience Act and its Technology Innovation and Emissions Reduction Regulation. It should be noted that these pieces of legislation do not explicitly reference data centres. Rather, the use of land and water by data centres, emissions by data centres, and construction of co-located power plants can trigger regulatory requirements. Depending on its design and operations, a particular data centre may not trigger all these regulatory processes but access to large amounts of electricity, water and land are key considerations for all data centres.

Electrical Supply If the data centre is designed to draw its electrical power from the grid, then it must obtain approval to do so from the Alberta Electric System Operator (AESO). If new transmission lines or substation upgrades are necessary, then the AESO will submit a Needs Identification Document application to the Alberta Utilities Commission (AUC) for approval. As well, an interconnection order also must be granted by the AUC to the local transmission facility owner for the physical connection to the grid. In anticipation of high demands from data centre proposals, the AESO recently reviewed the capacity of the grid and determined that a maximum of 1,200 MW could be allocated to data centres. That allocation has already been made (to Greenlight Electricity Centre/GLDC Load and Keephills Data Centre Phase 1) and any remaining data centre proposals are in a queue awaiting more capacity to be developed. The Data Centre Regulation – promulgated under the Electric Utilities Act – came into force on June 9th. This regulation provides a framework for “large” data centres (i.e. maximum demand greater than 75 MW) and directs the AESO to prioritize large data centres that bring their own power as opposed to those connecting to the electrical grid. This priority is accompanied by a new levy imposed on large grid-connected data centres via the Alberta Corporate Tax Act (Part 9.1) and a corresponding tax credit. The Data Centre Regulation also enables the AESO to set connection criteria for data centres seeking access to the grid. On June 12th, the AESO published its Guide to AESO Connection Requirements for Transmission Connected Data Centres which outlines the obligations to be met by data centre developers to ensure system reliability and efficient grid integration. As an alternative to grid connection, a data centre may choose to bring its own power (note that there usually is still a grid connection for redundancy purposes but the main source of electricity is provided by the data centre developer). This can be achieved by the data centre developer entering into a power purchase agreement or building its own power plant. The construction and operation of power plants is regulated by the AUC pursuant to the Hydro and Electric

Energy Act (the only exception being nuclear facilities which are regulated by the federal Canadian Nuclear Safety Commission). Power plant applications must meet the requirements as set out in Rule 007: Facility Applications and Rule 012: Noise Control. It should be noted that there are some limited exemptions from the requirement to file an application for a power plant approval (see section 4.1 of Rule 007) but, even in these cases, the AUC has jurisdiction to investigate compliance issues with Rules 007 and 012. Municipal Permits and Zoning Under the Municipal Government Act (MGA), municipalities have control over development within their boundaries. Under the MGA, municipalities must have a Land Use Bylaw (LUB) which sets out their development permit process and zoning requirements, and the LUB will guide the municipal process for permitting data centres. It is likely that new municipal plans and zoning amendments will be required to enable the construction and operation of a data centre (see for example, the Beacon AI Hub Area Structure Plan). In addition, a data centre is required to obtain development and building permits from the municipality. These permits are very likely to be subject to conditions designed to address municipal concerns around the data centre construction and operations. It should be noted that if a co-located power plant is approved by the AUC, section 619 of the MGA provides that the AUC approval supersedes any municipal land use bylaw or statutory plans. Water Needs A data centre may trigger both approval and licence requirements under the Water Act. An approval is required for activities that may alter the flow or level of water, may change the location or direction of flow of water, may cause the siltation of water or the erosion of any bed or shore of a water body, or may cause an effect on the aquatic environment (section 1(1)(b)). As an example, this could include constructing a data centre that disturbs a wetland (note that this also would trigger the Wetland Policy). A water licence is required for a diversion of water (which is defined in section 1(1)(m) as “the impoundment, storage, consumption, taking or removal of water for any purpose”). Data centres typically rely on water for the purposes of cooling computer equipment (either directly within the data centre as a heat transfer medium or indirectly by using additional electrical power for cooling purposes). Although it seems that necessary licenses should be obtained by the data centre owner/operator, we note that at least in the case of the proposed Wonder Valley AI Data Centre, the preliminary certificate for a licence to divert up to 24,000,000 cubic metres of water annually has been granted to the municipality for industrial and commercial purposes within the Greenview Industrial Gateway (see the Preliminary Certificate). Substance Releases and Clean-Up The EPEA regulates activities that may impact the environment using a system of approvals, registrations and notices (i.e. statutory authorization). The activities to which EPEA applies are listed in the Act’s Schedule of Activities and the type of statutory authorization required is set out in the Activities Designation Regulation. The Schedule does not include data centres, and so no approval, registration or notice is required for the data centre per se. However, the Schedule does include the release of substances that may cause an adverse effect; power plants (thermal, hydro-electric, solar and wind); and any activity, diversion of water, operation of a works or transfer of an allocation of water under a licence for which an approval, license or an approval of a transfer of an allocation of water under the Water Act is required. This would bring at least some aspects of a data centre development into the purview of EPEA. In addition, section 109 of EPEA prohibits the release of substances into the environment in “an amount, concentration or level or at a rate of release that causes or may cause a significant adverse effect” in circumstances where an approval, registration or notice is not required. Data centres may release air or water emissions as a result of their operations and those emissions must not contravene the requirements of section 109. Furthermore, the conservation and reclamation provisions in Part 6 of EPEA and the Conservation and Reclamation Regulation apply to power plants (thermal, hydro-electric, solar, wind and geothermal). These provisions require land impacted by these activities to be returned to equivalent land capability (i.e. land uses after conservation and reclamation are to be similar to those prior to the activity being conducted but not necessarily identical). Greenhouse Gas Emissions Much like data centre regulation, the regulation of greenhouse gas (GHG) emissions in Alberta is in the process of evolution. Provincially, GHG emissions are regulated by the Emissions Management and Climate Resilience

Act and its Technology Innovation and Emissions Reduction Regulation (TIER). These pieces of legislation set out the provincial carbon pricing system and require certain facilities to reduce their GHG emissions intensity (i.e. the amount of GHG released in production of a product using benchmarked units). Large emitters – i.e. facilities that release 100,000 CO₂e tonnes or import more than 10,000 tonnes of hydrogen annually – must participate in the TIER system. Other emitters may choose to participate. A facility is defined as a plant, structure or site (or any 2 or more contiguous plants, structures or sites) where one or more activities listed in the EPEA Schedule of Activities occur (other than a release of substance that may cause an adverse effect). As mentioned, data centres per se do not appear in the Schedule of Activities but power plants, and activities and diversions of water with Water Act authorizations do fall into the Schedule of Activities. The TIER contains some specific provisions related to electricity facilities. Depending on the scale of emissions from a data centre co-located power plant, TIER requirements may be applicable. The federal Greenhouse Gas Pollution Pricing Act and the Clean Electricity Regulations (made under the Canada Environmental Protection Act, 1999) are also relevant to GHG emissions. The Greenhouse Gas Pollution Pricing Act is designed to set a national carbon price and acts as a back-up in the event provincial pricing is insufficient. The Clean Electricity Regulations regulate GHG emissions associated with fossil fuel-based electrical generation. Although there appears to be no formal order in place, the Canada-Alberta MOU and associated Implementation Agreement have resulted in commitments to certain levels of carbon pricing being implemented in Alberta via TIER and to place the Clean Energy Regulations in abeyance. What about environmental assessments? Provincially, the environmental assessment (EA) process is governed by EPEA and applies to activities as designated in the Environmental Assessment (Mandatory and Exempted Activities) Regulation. Some activities listed in the EPEA Schedule are required to undergo EA, some are exempt from EA, and others are subject to EA at the discretion of a Director or the Minister (sections 41 and 47). EAs are mandatory for power plants that use non-gaseous fuel with a capacity of 100 MW or larger (i.e. EA is not mandatory for natural gas power plants) and for some large water diversion projects. Exemptions from EA exist for backup power generating systems that are only used to provide power when the principle source of power is interrupted and that is not connected to the grid, and for solar and wind power plants smaller than 1 MW (although the Minister may order an EA nonetheless). As mentioned, data centres are not included in EPEA's Schedule of Activities per se but power plants and activities requiring a Water Act license or approval are included, and this should bring data centre projects into the scope of the provincial EA process. In recent months, at least two proposed projects have been determined to not require provincial EAs: the Wonder Valley Project and the Synapse Data Center Project. In both cases, the Director indicated that a data centre is not a mandatory activity for the purposes of EA and considering the factors listed in section 44(3) of EPEA, further assessment was not required. The factors listed in section 44(3) are: - the location, size and nature of the proposed activity; - the complexity of the proposed activity and the technology to be employed in it; - any concerns in respect of the proposed activity that have been expressed by the public of which the Director is aware; - the presence of other similar activities in the same general area; - any other criteria established in the regulations; and - any other factors the Director considers to be relevant. The Director did reserve the ability to review the decision in the event of new information coming to light and noted that under section 47 of EPEA, the Minister could require an EA Report be prepared despite the Director's decision. There is also a federal impact assessment (IA) process under the Impact Assessment Act (IAA). If a proposed project falls into the list found in the Physical Activities Regulations, then a federal IA may be required. The project list includes large fossil-fuel and hydro-electric power plants and certain water projects. In order for a federal IA to proceed, there must be some effect on areas of federal jurisdiction such as impacts on fish and fish habitat, species at risk, migratory birds, and Indigenous peoples. If a proposed project is not on the project list, then the Minister may designate it to undergo an IA where there may be adverse effects within federal jurisdiction (IAA, section 9) At least 2 proposed data centres in Alberta have been deemed exempt from the federal IA process: the Mihta Askiy Data Centre and the Beacon AI Centers

Heartland Project. In both cases, the Impact Assessment Agency of Canada determined that, although the projects may cause adverse effects within federal jurisdiction, those effects would be addressed through existing federal and provincial legislation. Moving forward with data centres The GOA has been clear that the development of data centres is a priority. No doubt this is driven by a desire to diversify the economy, use Alberta's abundant natural gas resources to generate electricity, create new municipal tax bases, and to create jobs. But in the rush to be a leader in large data centre development, we should not become leaders in environmental damage. While there have been recent regulatory and policy developments to address the impacts of large data centres on the electrical grid, these changes are not enough. Other data centre impacts need to be addressed. This means including data centres as activities under EPEA to require statutory authorizations under that Act; and performing EAs so that the impacts on land, water and air are considered, and to ensure sufficient reclamation of decommissioned sites. In addition, data centres should not be allowed to piggyback on municipal water licences (as seems to be the case with Wonder Valley). Data centre developers should be required to obtain their own water licence to ensure transparency and accountability. Without robust EA and approval and licensing requirements in place, we are moving ahead with data centres with little monitoring and data on their environmental impact. Even though the work of data centres may be virtual, the impacts are very real: use of tremendous amounts of electricity, water and land. And, as mentioned, there is often local opposition driven by concerns around water impacts, land use (loss of agricultural lands), local heat island impacts, noise and visual impacts. These concerns need to be heard and addressed. [1] This definition is also adopted in the Guide to AESO Connection Requirements for Transmission Connected Data Centres at 4. The Alberta Corporate Tax Act uses a very similar definition (section 92.2(h)). [2] Note that the MOU was followed by an implementation agreement which set out joint action to strengthen carbon markets, double the electricity grid and diversify global markets for energy projects but made no direct reference to data centres. THANKS FOR YOUR SUPPORT Your support is vital for stronger environmental legislation. As Alberta's leading environmental charity, the Environmental Law Centre has served our community for over 40 years, providing objective guidance on crucial legislative changes. Your contribution helps protect our environment for future generations. Please support our work: Share, engage and donate to the ELC